

Strategic Product Placement Analysis

Final Project Report

1.INTRODUCTION

1.1 Project Overview

Strategic Product Placement Analysis is a business intelligence project developed using Tableau to analyze the relationship between product positioning, sales performance, pricing, promotions, customer demographics, seasonal demand, and store foot traffic. The project uses a retail sales dataset to create interactive dashboards and visualizations that help identify patterns affecting product sales. By analyzing these factors, the project provides meaningful insights that support better product placement strategies and improve overall business performance. The dashboards enable retail managers and business analysts to make informed, data-driven decisions that enhance customer satisfaction and maximize revenue.

1.2 Purpose

The purpose of this project is to analyze retail sales data and understand how different business factors influence product sales and customer purchasing behavior. Using Tableau, the project visualizes sales trends, product placement effectiveness, pricing strategies, promotional impact, and customer demographics through interactive dashboards. The insights generated help retail businesses identify the most effective product positioning strategies, optimize marketing efforts, improve inventory planning, and make informed business decisions that contribute to increased sales and revenue growth.

2.IDEATION PHASE

2.1 Problem Statement

PS-1 : Product Placement Decision Making



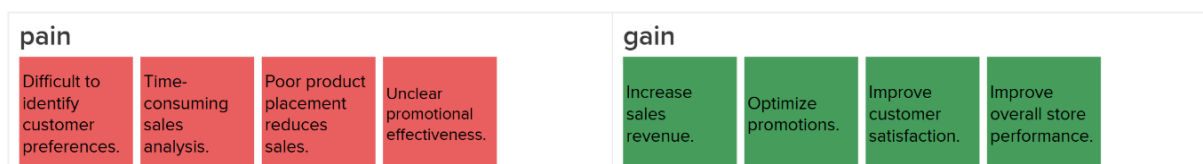
PS-2 : Marketing and Sales Optimization

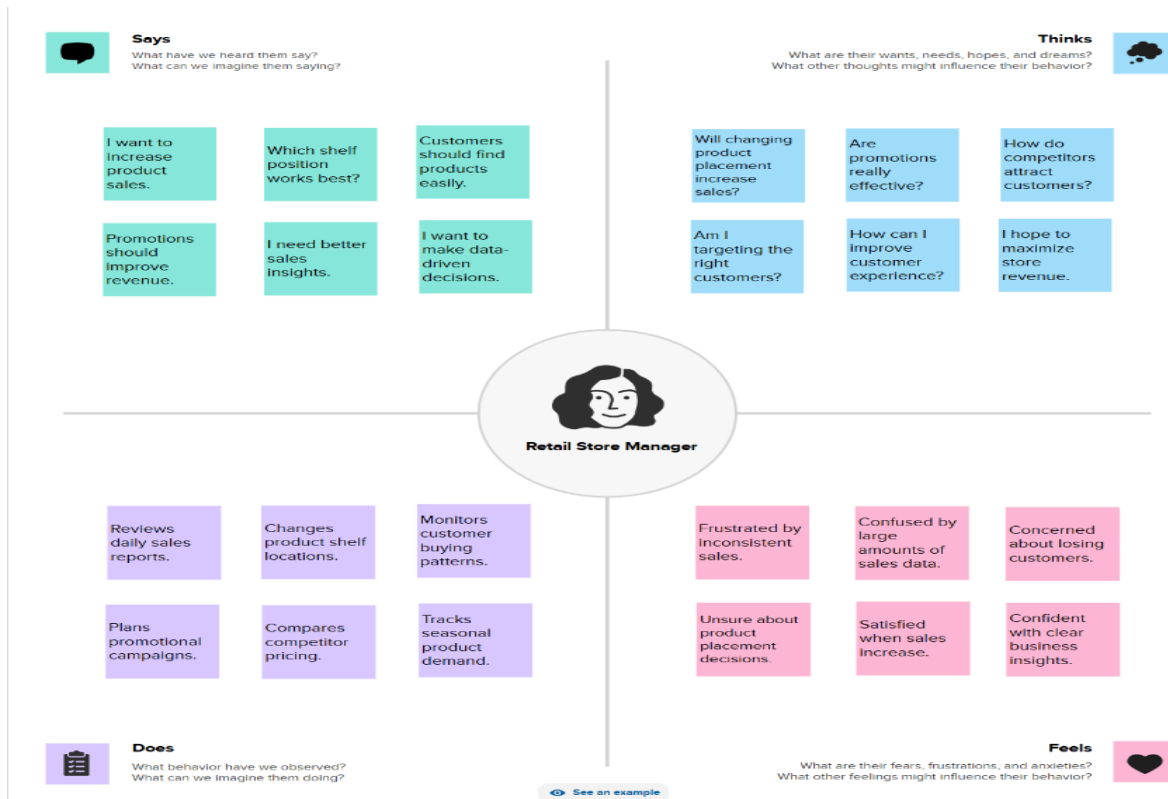


Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Retail Store Manager	Increase sales through better product placement	Cannot identify the best positions	Data is difficult to analyze	Uncertain
PS-2	A Retail Business Owner	Improve sales through better promotions and pricing.	I don't know which strategy works best.	I cannot clearly understand the sales data.	Confused about making business decisions.

2.2 Empathy Map Canvas

Empathy map for Strategic Product Placement Analysis



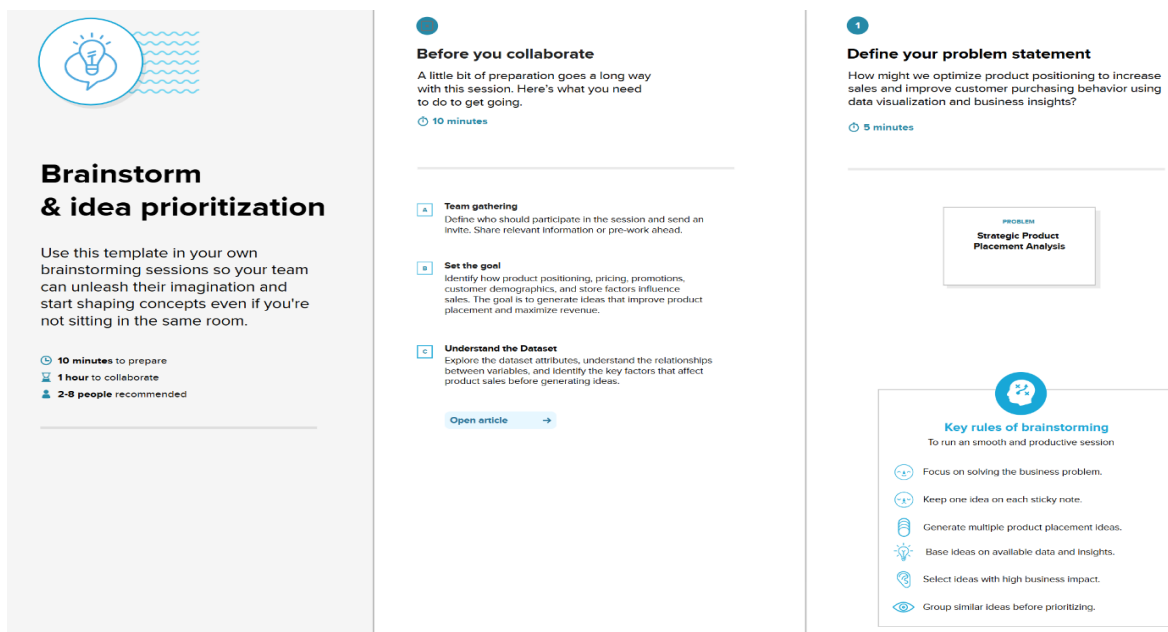


Reference:

<https://app.mural.co/t/jaswanth0805/m/jaswanth0805/1782550162805/40aa3eeae21b691cb5f31d6cd989d8ec0c76ae09?sender=u752610297b44303dfc182760>

2.3 Brainstorming

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step-2: Brainstorm, Idea Listing and Grouping

3 Brainstorm

Write down any ideas that come to mind that address your problem statement.

10 minutes

TIP You can select a sticky note and hit the pencil icon to start drawing!

Product Position
 Brand of Product
 Target Market
 Key Features
 Key Benefits
 Key Differentiators

Pricing
 Price Range
 Price Points
 Price Strategy
 Price Sensitivity
 Price Elasticity

Promotion
 Promotion Channels
 Promotion Types
 Promotion Timing
 Promotion Location
 Promotion Message

Consumer Demographics
 Age Group
 Gender
 Income Level
 Education Level
 Marital Status

Foot Traffic
 Foot Traffic Volume
 Foot Traffic Density
 Foot Traffic Duration
 Foot Traffic Frequency

Seasonal Demand
 Seasonal Demand Volume
 Seasonal Demand Density
 Seasonal Demand Duration
 Seasonal Demand Frequency

Product Category
 Product Category Volume
 Product Category Density
 Product Category Duration
 Product Category Frequency

Business Improvement
 Business Improvement Volume
 Business Improvement Density
 Business Improvement Duration
 Business Improvement Frequency

4 Group Ideas

Take turns sharing your ideas while clustering similar or related notes as you go. Once all sticky notes have been grouped, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.

20 minutes

TIP Add customizable tags to sticky notes to make it easier to find, browse, organize, and categorize important ideas as themes within your mind.

Product Placement
 Product Placement Volume
 Product Placement Density
 Product Placement Duration
 Product Placement Frequency

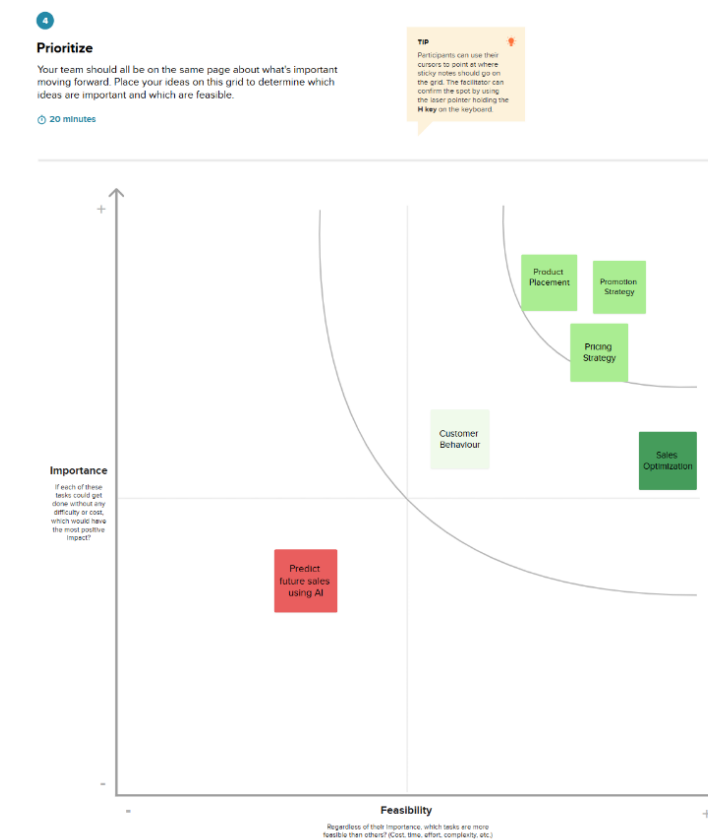
Pricing Strategy
 Pricing Strategy Volume
 Pricing Strategy Density
 Pricing Strategy Duration
 Pricing Strategy Frequency

Promotion Strategy
 Promotion Strategy Volume
 Promotion Strategy Density
 Promotion Strategy Duration
 Promotion Strategy Frequency

Customer Behaviour
 Customer Behaviour Volume
 Customer Behaviour Density
 Customer Behaviour Duration
 Customer Behaviour Frequency

Sales Optimization
 Sales Optimization Volume
 Sales Optimization Density
 Sales Optimization Duration
 Sales Optimization Frequency

Step-3: Idea Prioritization



Reference:

<https://app.mural.co/t/jaswanth0805/m/jaswanth0805/1782480227433/6f18abd6b4b5e15a018bb6cf010991759e9b2176?sender=u752610297b44303dfc182760>

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map



3.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Import & Preparation	Import retail sales dataset into Tableau. Verify data types and field names. Clean and prepare data for visualization.
FR-2	Sales Analysis Dashboard	Display total sales volume. Show sales by product category. Compare sales across product positions.
FR-3	Product Placement Analysis	Compare Front of Store, End-cap, and Aisle positions. Identify high-performing product locations. Analyze the relationship between placement and sales.
FR-4	Customer & Promotion Analysis	Analyze customer demographics.

		Compare promotional and non-promotional sales. Study the impact of foot traffic.
FR-5	Interactive Dashboard	Apply filters for Product Category. Apply filters for Promotion. Apply filters for Foot Traffic.
FR-6	Business Insights	Generate actionable insights. Identify best-performing products. Recommend pricing and promotion strategies.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

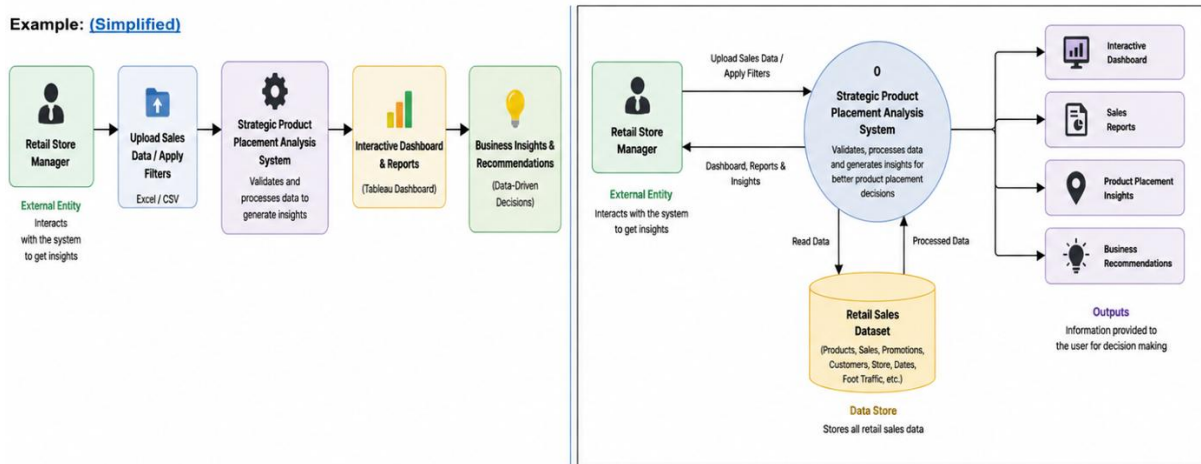
FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The Tableau dashboard should be simple, interactive, and easy for retail managers to use.
NFR-2	Security	Only authorized users should access the business dashboard and sales data.
NFR-3	Reliability	The dashboard should consistently display accurate and validated sales information.
NFR-4	Performance	Charts, filters, and dashboards should load quickly and respond without noticeable delay.
NFR-5	Availability	The dashboard should be accessible whenever required for business analysis.
NFR-6	Scalability	The solution should support larger datasets and additional retail branches without major redesign.

3.3 Data Flow Diagram

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: (Simplified)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Retail Store Manager	Data Import	USN-1	As a Retail Store Manager, I want to import the retail sales dataset so that I can analyze sales performance.	Dataset is imported successfully and ready for analysis.	High	Sprint-1
	Dashboard	USN-2	As a Retail Store Manager, I want to view an interactive dashboard so that I can monitor key sales metrics.	Dashboard displays KPIs, charts, and sales summaries correctly.	High	Sprint-1
	Product Placement Analysis	USN-3	As a Retail Store Manager, I want to compare product positions so that I can identify the best-performing shelf locations.	Product position comparison is displayed accurately.	High	Sprint-1
	Promotion Analysis	USN-4	As a Retail Store Manager, I want to analyze promotional sales so that I can evaluate promotion effectiveness.	Promotion-wise sales charts are available.	High	Sprint-2
	Interactive Filters	USN-5	As a Retail Store Manager, I want to filter the dashboard by category, promotion, foot traffic, and season so that I can perform detailed analysis.	Filters update all dashboard visualizations correctly.	High	Sprint-2
Business Analyst	Customer Analysis	USN-6	As a Business Analyst, I want to analyze customer demographics so that I can understand purchasing behavior.	Dashboard displays demographic-based analysis.	Medium	Sprint-2
	Pricing Analysis	USN-7	As a Business Analyst, I want to compare product prices with competitor prices so that I can evaluate pricing strategies.	Price comparison charts are displayed correctly.	Medium	Sprint-2
	Business Insights	USN-8	As a Business Analyst, I want to generate business insights and recommendations so that I can support management decisions.	Dashboard provides actionable recommendations based on analysis.	High	Sprint-3
	Report Generation	USN-9	As a Business Analyst, I want to export reports and dashboards so that I can present findings to stakeholders.	Reports can be exported in the required format.	Medium	Sprint-3
	Performance Monitoring	USN-10	As a Business Analyst, I want to monitor sales trends over time so that I can identify growth opportunities.	Trend analysis charts are available and accurate.	Medium	Sprint-3

3.4 Technology Stack

Guidelines:

- ☐ Import and validate the retail sales dataset before analysis.
- ☐ Clean and preprocess the data to ensure accuracy and consistency.
- ☐ Build interactive Tableau dashboards with filters and visualizations.
- ☐ Analyze product positioning, pricing, promotions, and customer behavior.

Technical Architecture

Strategic Product Placement Analysis System

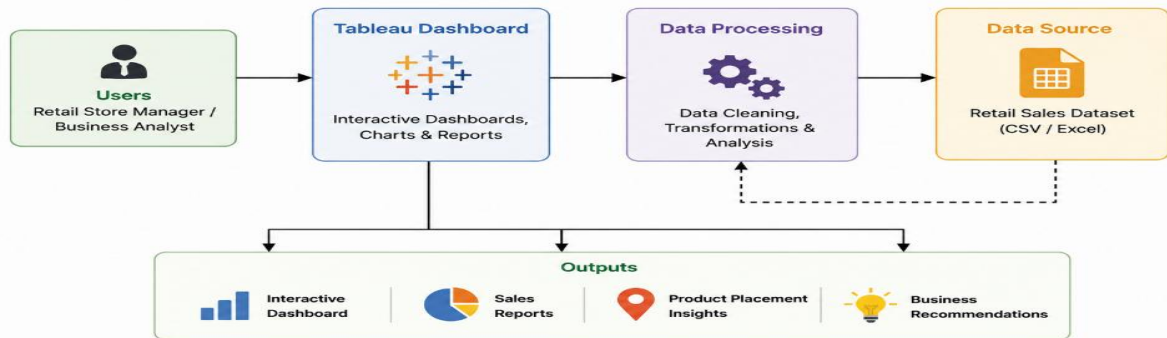


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interactive dashboard for business users	Tableau Desktop
2.	Data Import	Import retail sales dataset	CSV / Excel
3.	Data Preparation	Clean and organize the dataset	Tableau Prep / Tableau Data Source
4.	Data Analysis	Analyze sales, pricing, promotions, and customer behavior	Tableau Calculated Fields
5.	Visualization	Interactive charts and dashboards	Tableau Desktop
6.	Data Storage	Store retail sales dataset	Local CSV File
7.	File Storage	Store project files and dashboard	Local File System
8.	External API-1	Not Applicable	Not Applicable
9.	External API-2	Not Applicable	Not Applicable
10.	Machine Learning Model	Not Used	Not Applicable
11.	Infrastructure (Server / Cloud)	Local Computer	Windows PC with Tableau Desktop

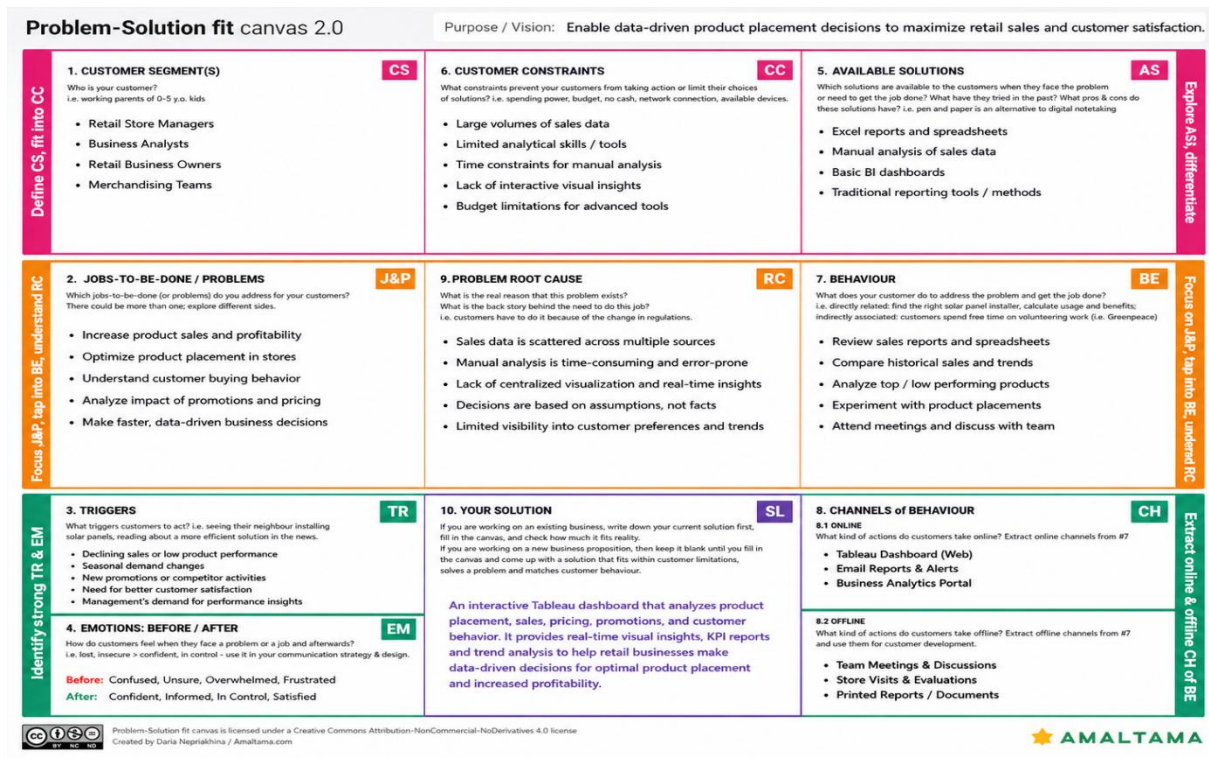
Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Dataset handled using Tableau features	Tableau Desktop
2.	Security Implementations	Dataset stored locally and accessed only by authorized users	Local File Security
3.	Scalable Architecture	Dashboard can analyze larger retail datasets by refreshing the data source	Tableau Data Source

S.No	Characteristics	Description	Technology
4.	Availability	Dashboard can be opened anytime on the local system	Tableau Desktop
5.	Performance	Optimized visualizations, filters, and calculated fields provide fast analysis	Tableau Optimization

4. PROJECT DESIGN

4.1 Problem Solution Fit



References:

- <https://www.ideahackers.network/problem-solution-fit-canvas/>
- <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>

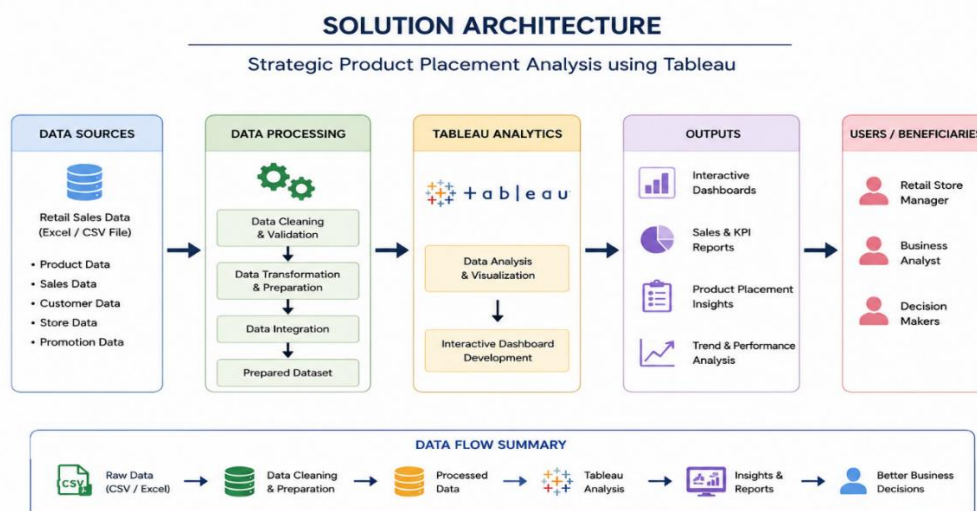
4.2 Proposed Solution

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Retail businesses often struggle to identify the most effective product placement strategies due to large volumes of sales data and limited analytical insights.
2.	Idea / Solution description	Develop an interactive Tableau dashboard that analyzes retail sales data to evaluate product positioning, pricing, promotions, customer demographics, foot traffic, and seasonal trends. The dashboard provides visual insights and recommendations to help managers make informed decisions that improve product placement and increase sales.

3.	Novelty / Uniqueness	The solution combines multiple business factors such as product placement, promotions, pricing, customer demographics, and seasonal demand into a single interactive Tableau dashboard. It enables easy visualization, filtering, and comparison of retail performance, helping businesses identify the most effective placement strategies.
4.	Social Impact / Customer Satisfaction	Better product placement improves product availability and the shopping experience, allowing customers to find products more easily. Businesses benefit from increased sales, while customers enjoy a more organized store layout and personalized product offerings.
5.	Business Model (Revenue Model)	The solution can be offered as a retail business analytics dashboard to supermarkets, retail chains, and shopping stores. Revenue can be generated through dashboard implementation, customization services, analytics consulting, training, and annual maintenance and support.
6.	Scalability of the Solution	The dashboard is scalable and can be used across multiple retail stores and branches. It supports larger datasets, additional product categories, and future integration with cloud databases, live sales systems, or machine learning models for predictive analytics.

4.3 Solution Architecture

Solution Architecture Diagram:



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

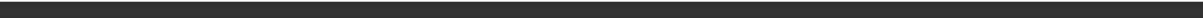
Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story No	User Story / Task	Story Points	Priority	Team Member
Sprint-1	Data Import	USN-1	As a Retail Store Manager, I want to import the retail sales dataset so that I can analyze product performance.	3	High	Member-1
Sprint-1	Data Preparation	USN-2	As a Retail Store Manager, I want the data to be cleaned and validated before analysis.	2	High	Member-2
Sprint-1	Dashboard Development	USN-3	As a Retail Store Manager, I want an interactive dashboard to monitor sales KPIs.	5	High	Member-3
Sprint-2	Product Placement Analysis	USN-4	As a Retail Store Manager, I want to compare product placements to identify the best-performing locations.	5	High	Member-1
Sprint-2	Customer Analysis	USN-5	As a Business Analyst, I want to analyze customer demographics and buying behavior.	3	Medium	Member-2
Sprint-2	Promotion Analysis	USN-6	As a Business Analyst, I want to evaluate promotion effectiveness.	3	Medium	Member-3

Sprint	Functional Requirement (Epic)	User Story No	User Story / Task	Story Points	Priority	Team Member
Sprint-3	Interactive Filters	USN-7	As a Retail Store Manager, I want dashboard filters for category, season, and promotion.	3	High	Member-4
Sprint-3	Business Insights	USN-8	As a Business Analyst, I want actionable recommendations for better product placement.	5	High	Member-1
Sprint-3	Report Export	USN-9	As a Business Analyst, I want to export dashboard reports for presentations.	2	Medium	Member-2
Sprint-4	Dashboard Testing	USN-10	As a Retail Store Manager, I want the dashboard to display accurate insights without errors.	3	High	Member-3
Sprint-4	Documentation	USN-11	As a Project Team, we want complete project documentation.	2	Medium	Member-4
Sprint-4	Final Deployment	USN-12	As a Retail Store Manager, I want the final dashboard ready for business use.	4	High	Member-1

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date
Sprint-1	10	6 Days	01 Jul 2026	06 Jul 2026	10	06 Jul 2026



Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date
Sprint-2	11	6 Days	07 Jul 2026	12 Jul 2026	11	12 Jul 2026
Sprint-3	10	6 Days	13 Jul 2026	18 Jul 2026	10	18 Jul 2026
Sprint-4	9	6 Days	19 Jul 2026	24 Jul 2026	9	24 Jul 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = 10 \div 6 = 1.67 \text{ Story Points / Day}$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

Day	Remaining Story Points
0	10
1	9
2	7
3	5
4	3
5	1
6	0

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

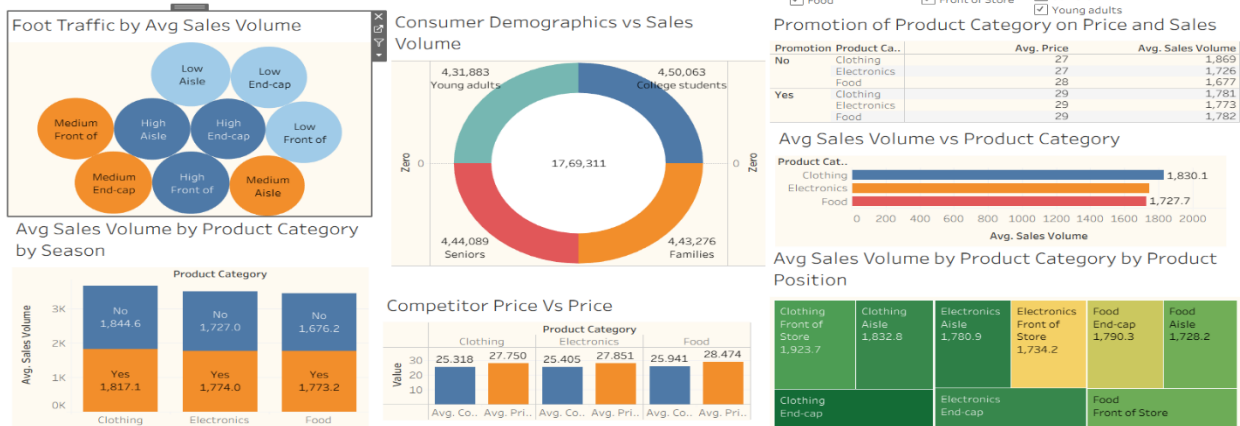
S.No.	Parameter	Screenshot / Values
1.	Data Rendered	We took a csv file named product positioning it is of 68kb. It contains 1000 rows and 10 columns.
2.	Data Preprocessing	We got the table clean there is no requirement to do it but we checked.
3.	Utilization of Filters	Applied interactive filters such as Product Category, Season, Promotion, Product Position, and Consumer Demographics to enable dynamic analysis.
4.	Calculation fields Used	Created calculated fields including Average Sales Volume, Average Price, Sales KPIs, and other aggregated measures used in dashboard visualizations.
5.	Dashboard design	No of Visualizations / Graphs - total 8 visualizations Avg Sales Volume vs Product Category Avg Sales Volume by Product Category by Product Position Avg Sales Volume by Product Category by Season Competitor Price Vs Price Consumer Demographics vs Sales Volume Foot Traffic by Avg Sales Volume Product Category vs Price Promotion of Product Category on Price and Sales Volume
6	Story Design	No of Visualizations / Graphs – used 6 visualizations Avg Sales Volume vs Product Category Avg Sales Volume by Product Category by Product Position Competitor Price Vs Price Consumer Demographics vs Sales Volume Foot Traffic by Avg Sales Volume Promotion of Product Category on Price and Sales Volume

7. RESULTS

7.1 Output Screenshots

Dashboard 1:

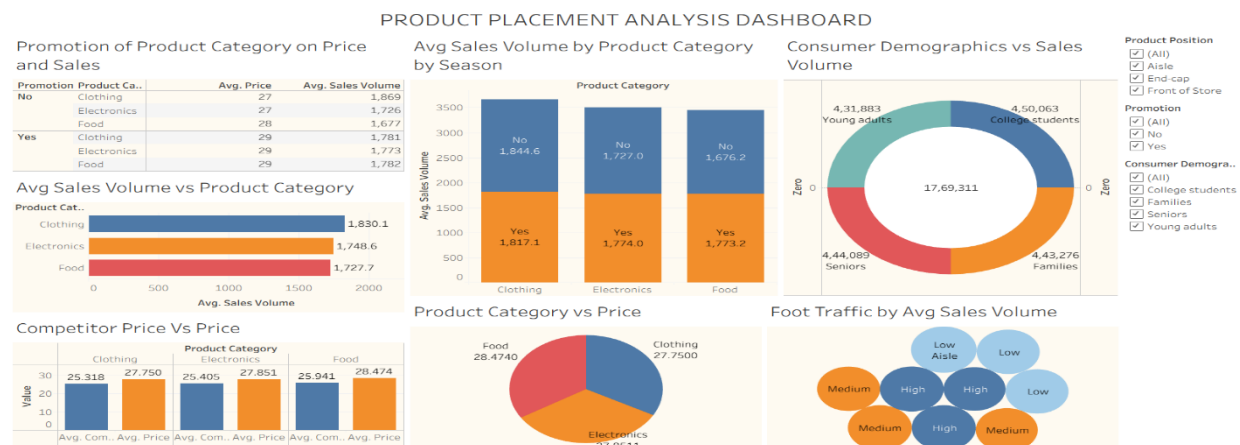
PRODUCT PLACEMENT ANALYSIS-DASHBOARD



Reference link:

https://public.tableau.com/views/productplacementanalsysdashboard1/PRODUCTPLACEMENTANALYSISDASHBOARD?:language=enUS&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Dashboard 2:

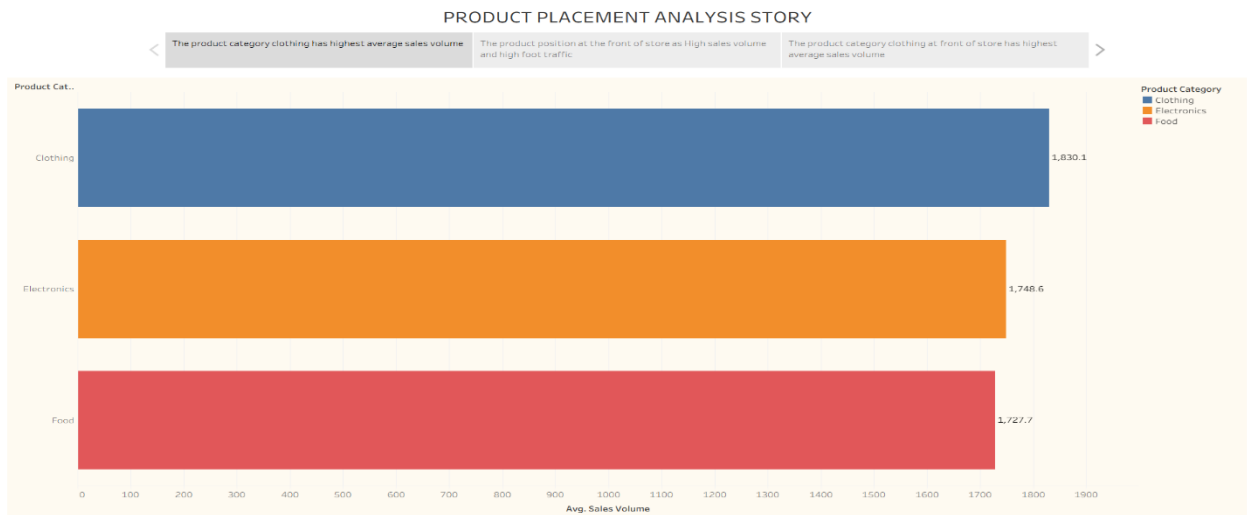


Reference link:

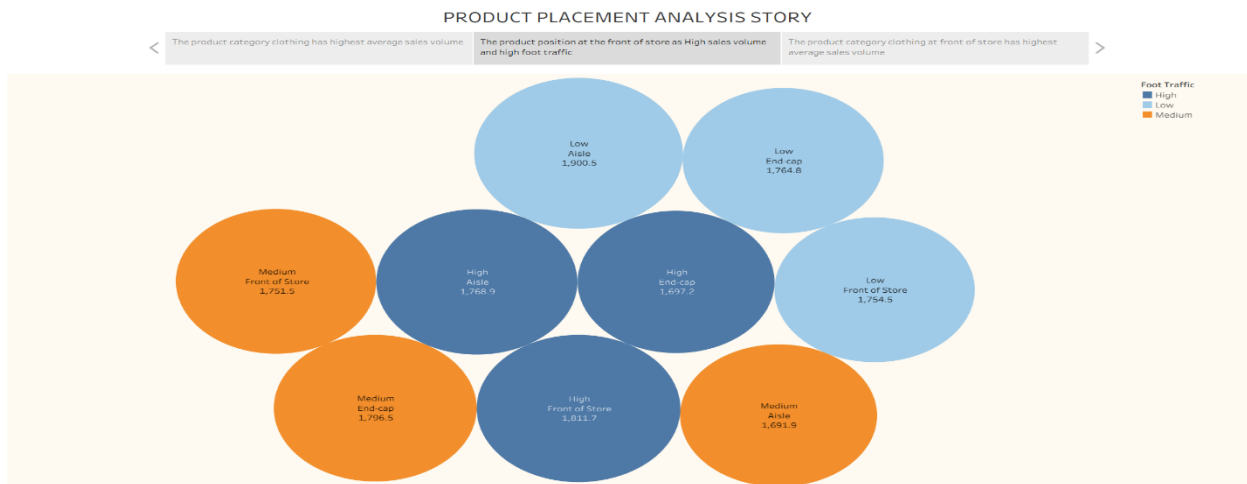
https://public.tableau.com/views/productplacementanalsysdashboard2/PRODUCTPLACEMENTANALYSISDASHBOARD?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Story:

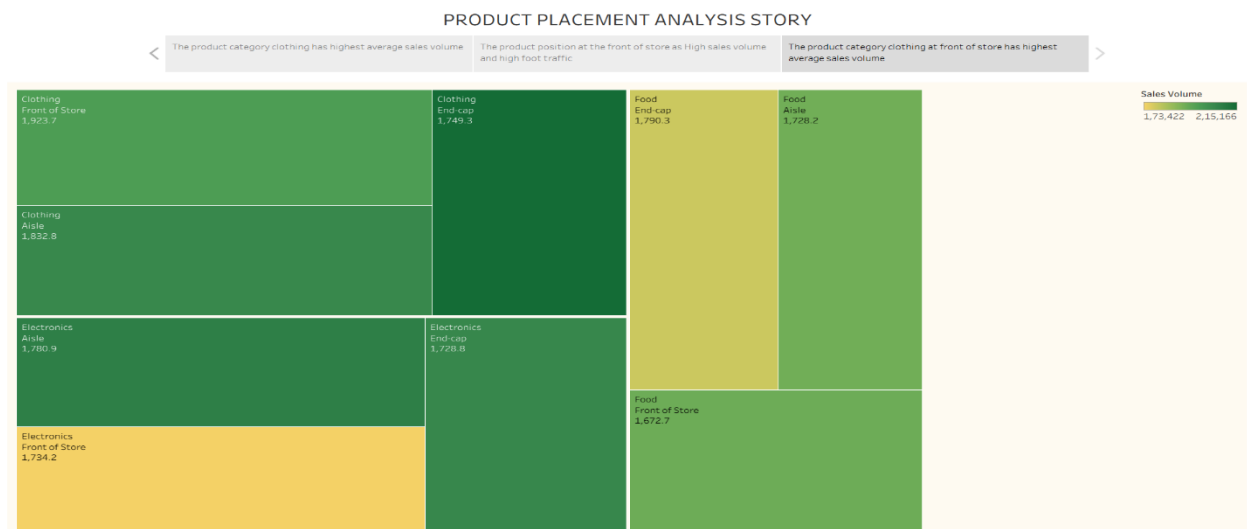
1. The product category Clothing has the highest average sales volume.



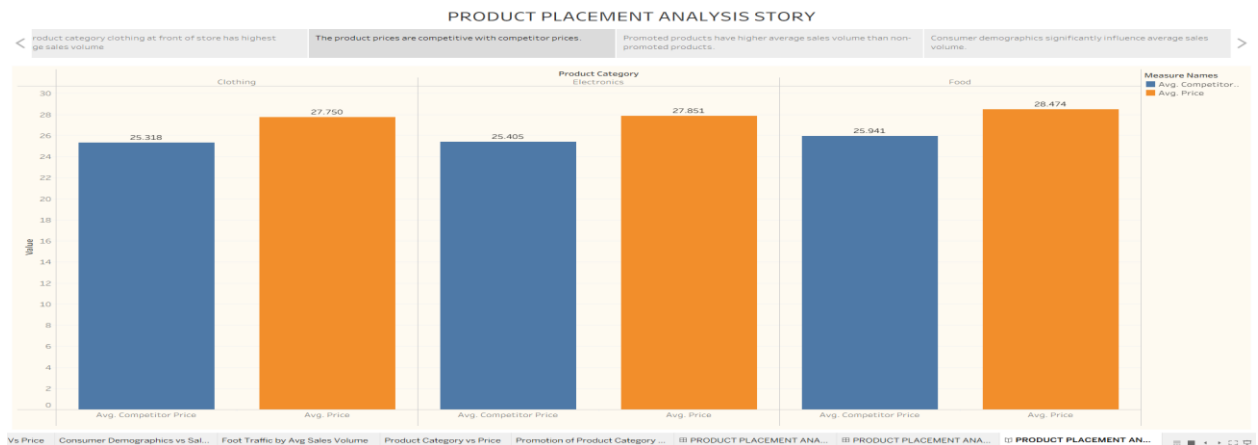
2. The product position at the Front of Store has high sales volume and high foot traffic.



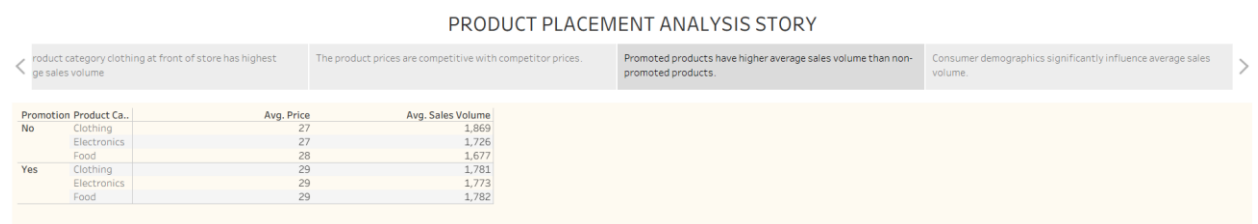
3. The product category Clothing at the Front of Store has the highest average sales volume.



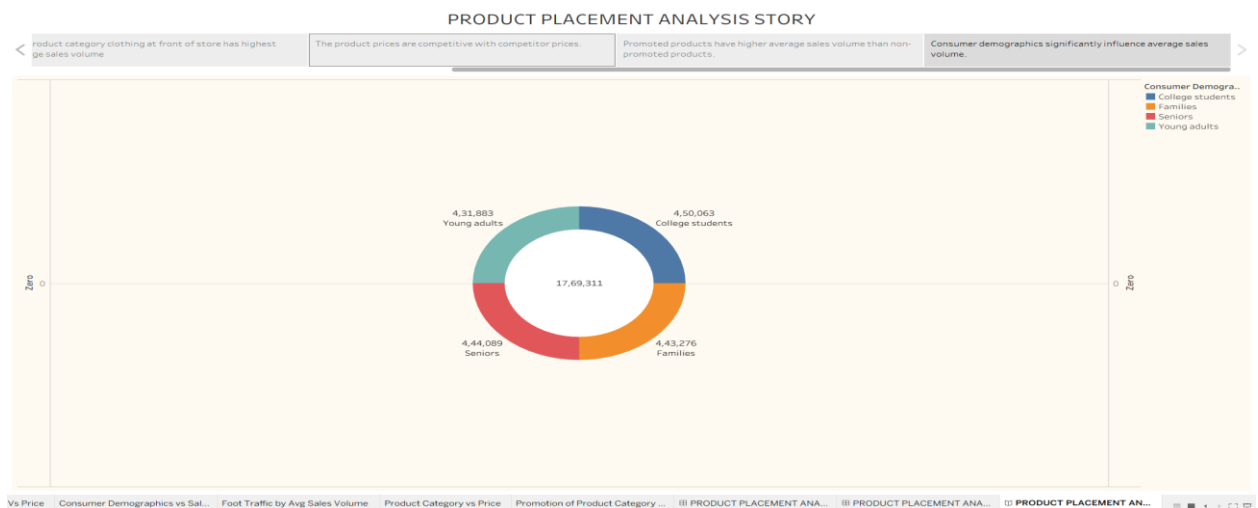
4. The product prices are competitive with competitor prices.



5. Promoted products have higher average sales volume than non-promoted products.



6. Consumer demographics significantly influence average sales volume.



Reference link:

https://public.tableau.com/views/Retail_Sales_Analysis_17823990226980/PRODUCTPLACEMENTANALYSISSTORY?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

8. ADVANTAGES & DISADVANTAGES

Advantages

1 Improves Product Placement Decisions

- Identifies the most effective product positions to maximize sales.

2 Supports Data-Driven Decision Making

- Helps managers make informed business decisions using visual insights.

3 Interactive Data Visualization

- Tableau dashboards make complex sales data easy to understand and analyze.

4 Enhances Sales Performance Analysis

- Compares sales across product categories, seasons, promotions, and customer demographics.

5 Evaluates Promotional Effectiveness

- Measures the impact of promotions on product sales and pricing.

6 Improves Customer Understanding

- Analyzes customer demographics and purchasing behavior to support targeted marketing.

7 Saves Time

- Reduces manual analysis through automated dashboards and interactive filters.

8 Easy to Customize

- Dashboards can be modified to include additional products, regions, or business metrics.

Disadvantages

1 Depends on Data Quality

- Incorrect or incomplete data can lead to inaccurate insights.

2 No Real-Time Data

- The analysis is based on historical data and does not reflect live sales updates.

3 Limited Predictive Capability

- The project focuses on descriptive analysis and does not forecast future sales.

4 Requires Tableau Software

- Users need Tableau Desktop or Tableau Public to view and interact with the dashboards.

5 Static Dataset

- The current dataset may not represent changing market conditions over time.

6 Limited External Factors

- Factors such as economic conditions, customer reviews, and competitor marketing are not included.

7 Learning Curve

- Users unfamiliar with Tableau may need basic training to use advanced dashboard features.

8 Scalability Depends on Resources

- Very large datasets may require better hardware or optimized Tableau extracts for smooth performance.

9. Conclusion

The **Strategic Product Placement Analysis** project successfully demonstrates how **Tableau** can be used to transform retail sales data into meaningful business insights. By analyzing product placement, sales performance, pricing, promotions, customer demographics, seasonal trends, and foot traffic, the project identifies the key factors that influence customer purchasing behavior and overall sales performance.

The interactive dashboards and story provide a clear understanding of sales patterns and reveal that product placement plays a significant role in improving sales. The analysis also highlights the impact of competitive pricing, promotional activities, and customer demographics on business performance. These insights enable retail managers and business analysts to make informed, data-driven decisions that optimize product positioning, improve marketing strategies, and enhance customer satisfaction.

Overall, this project achieves its objective of providing a practical and effective business intelligence solution for retail analysis. The developed Tableau dashboards serve as a valuable decision-support tool that helps retailers improve operational efficiency, maximize revenue, and strengthen their competitive advantage in the retail market.

10. Future scope

The **Strategic Product Placement Analysis** project can be further enhanced by integrating advanced technologies and expanding its analytical capabilities. Future improvements include:

1. Real-Time Data Integration

- Connect the dashboard to live retail databases for real-time sales monitoring and analysis.

2. Predictive Analytics

- Integrate machine learning models to forecast future sales trends and customer demand.

3. Cloud-Based Deployment

- Publish the dashboard on cloud platforms such as Tableau Server or Tableau Cloud for secure access from anywhere.

4. Automated Data Refresh

- Enable automatic data updates to eliminate manual data imports and ensure up-to-date insights.

5. **Mobile Dashboard Support**

- Develop mobile-friendly dashboards for easy access by managers and decision-makers on smartphones and tablets.

6. **Advanced Customer Segmentation**

- Incorporate additional customer attributes such as purchase history, loyalty programs, and shopping preferences for deeper analysis.

7. **AI-Based Business Recommendations**

- Implement artificial intelligence to automatically suggest optimal product placement, pricing strategies, and promotional campaigns.

8. **Multi-Store and Multi-Region Analysis**

- Extend the solution to analyze data from multiple stores and regions, enabling comparative performance analysis and enterprise-wide decision making.

11. Appendix

Dataset link:

<https://www.kaggle.com/datasets/amitvkulkarni/impact-of-product-positioning-on-sales>

Git-hub link:

<https://github.com/KadimisettiJaswanth/Strategic-Product-Placement-Analysis.git>

Project demo link:

<https://drive.google.com/file/d/1wIXLMkTJ3jPbs97ZyNs9eK2OshkHldXy/view?usp=sharing>